

Rapid fabrication of zwitterionic coating on 316L stainless steel surface for marine biofouling resistance

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ABSTRACT

Zwitterionic polymers are considered as nontoxic fouling-resistant materials, and have been widely studied in recent years. However, the immobilization of zwitterionic polymers on metal surfaces still suffers from the complicated and time-consuming procedures. In this work, a zwitterionic polydopamine/poly(sulfobetaine methacrylate) (PDA/PSB) coating on 316L stainless steel surface was fabricated by simply immersing in a mixed solution of dopamine, sulfobetaine methacrylate (SB) and oxidants for 30 mins. The anti-fouling performance of PDA/PSB coating was evaluated in terms of organic molecules (protein and polysaccharide) adsorption and microbes (Gram-negative bacterium, Gram-positive bacterium and diatom) adhesion. The results showed that the PDA/PSB coating inhibits the adsorption of BSA and sodium alginate, and further reduces the adhesion of *Pseudomonas aeruginosa*, *Bacillus subtilis* and *Navicula* sp. by more than 99% in 24 h. This research provides an important inspiration for the practical application of zwitterionic polymers in the field of marine anti-fouling.

1. Introduction

Marine biofouling refers to a phenomenon that the artificial surface in seawater is subjected to marine organisms' colonization, which has been a worldwide problem for decades [1]. This kind of undesired colonization has a series of adverse effects, such as increasing the sailing resistance of vessels, accelerating the corrosion of marine infrastructures, and blocking seawater pipelines, etc. According to a recent report, just in the shipping industry, the global economic loss caused by biofouling is more than 150 billion USD per year [2]. Therefore, there is a great demand to develop effective methods for preventing or reducing marine biofouling. Generally, there are several key steps involved in marine biofouling [3–5]. Firstly, the organic molecules, i.e. proteins and polysaccharides, in seawater physically adsorb on the surfaces to form a conditioning film. Then, bacteria adhere on the surface and gradually develop into a biofilm by secreting adhesive extracellular polymeric substances (EPS). Afterwards, diatoms and spores of microalgae accumulate to form a slime layer, i.e. micro-fouling. Finally, marine macro-organisms like barnacles grow and develop into macro-fouling. It has been widely accepted that preventing the adsorption of organic molecules and the attachment of microorganisms at the early stage is essential to control the development of marine biofouling [6].

Zwitterionic polymers, such as poly(sulfobetaine methacrylate)

(PSB) and poly(carboxybetaine methacrylate) (PCB), are considered to be one of the most promising non-fouling materials [7–9]. They have exhibited a long-term stability even in complex media, such as blood plasma and serum [10]. Previous studies have confirmed that the surfaces coated with zwitterionic polymers can resist nonspecific protein adsorption and bacterial adhesion [11,12]. The key factor for anti-fouling property of zwitterionic polymers is the formation of a strong hydration layer due to the presence of the mixed anionic and cationic groups in their molecular structure [13]. Currently, anchoring zwitterionic polymers onto a surface is usually achieved by “graft-from” and “graft-to” techniques [14–17]. The “graft-from” method involves the immobilization of initiator on the surface and the surface-initiated polymerization. For example, Zhang et al. [18,19] constructed uniform PSB and PCB brushes on the initiator-covered glass surface via atom transfer radical polymerization. The “graft-to” method involves the pre-preparation of zwitterionic polymers with adhesive groups. For instance, Sundaram et al. [17] prepared a zwitterionic sulfobetaine-based polymer with a catechol chain end, which was successfully grafted to a variety of surfaces, such as polystyrene, polyvinyl chloride and nylon, etc. However, there are some disadvantages during fabrication of the zwitterionic polymers onto materials surfaces using above two methods, i.e., the “graft from” method requires pre-treatment of the surface and an inert atmosphere during the polymerization, while the

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“graft-to” method suffers from the complex synthesis and purification process of polymers. It's still a great challenge to develop the simple and versatile approaches for anchoring zwitterionic polymers onto surfaces.

Mussel-inspired chemistry is one of the promising methods to break through the above limitations of surface modification [20]. Dopamine is a small-molecule compound with similar functionalities to 3,4-dihydroxy-L-phenylalanine (DOPA), which is a main component of adhesive proteins secreted by mussel. It can be oxidized and polymerize to form polydopamine (PDA) in a weakly alkaline aqueous medium [21,22]. Due to the presence of a large amount of catechol and amine groups, the PDA coating is able to be deposited on many types of organic and inorganic materials surfaces, and served as an intermediate layer for secondary modification [23–26]. For example, Cao et al. [27,28] used PDA coating as adhesive layer to immobilize antibacterial peptides on stainless steel (SS) surface to prevent bacterial adhesion. Qiao et al. [29] proposed a two-step method of sequential depositing PDA and PSB coatings on polypropylene meshes to enhance their biocompatibility. Although the strategy of two-step deposition is successful in fabricating

multifunctional coatings, it is still a time-consuming deposition process (about 24 h). Most recently, Zhang et al. [30] found that the free radicals generated during the oxidative polymerization of dopamine can trigger the polymerization of acrylate monomers, which provides the possibility for the facile and rapid fabrication of zwitterionic coatings. However, the exact mechanism of polymerization is still unclear, and the anti-fouling performance of the zwitterionic coating prepared by this method remains to be evaluated.

As a special marine engineering material, although SS has a reasonable corrosion resistance, it faces the problem of biofouling inevitably. Herein, marine biofouling on 316L SS surface was inhibited based on zwitterionic PDA/PSB coating, which was rapidly prepared through one-step simultaneous polymerization and codeposition of dopamine and SB. The anti-fouling performances of the PDA/PSB coating were evaluated by detecting the adsorption of organic molecules and the adhesion of bacteria and diatoms.

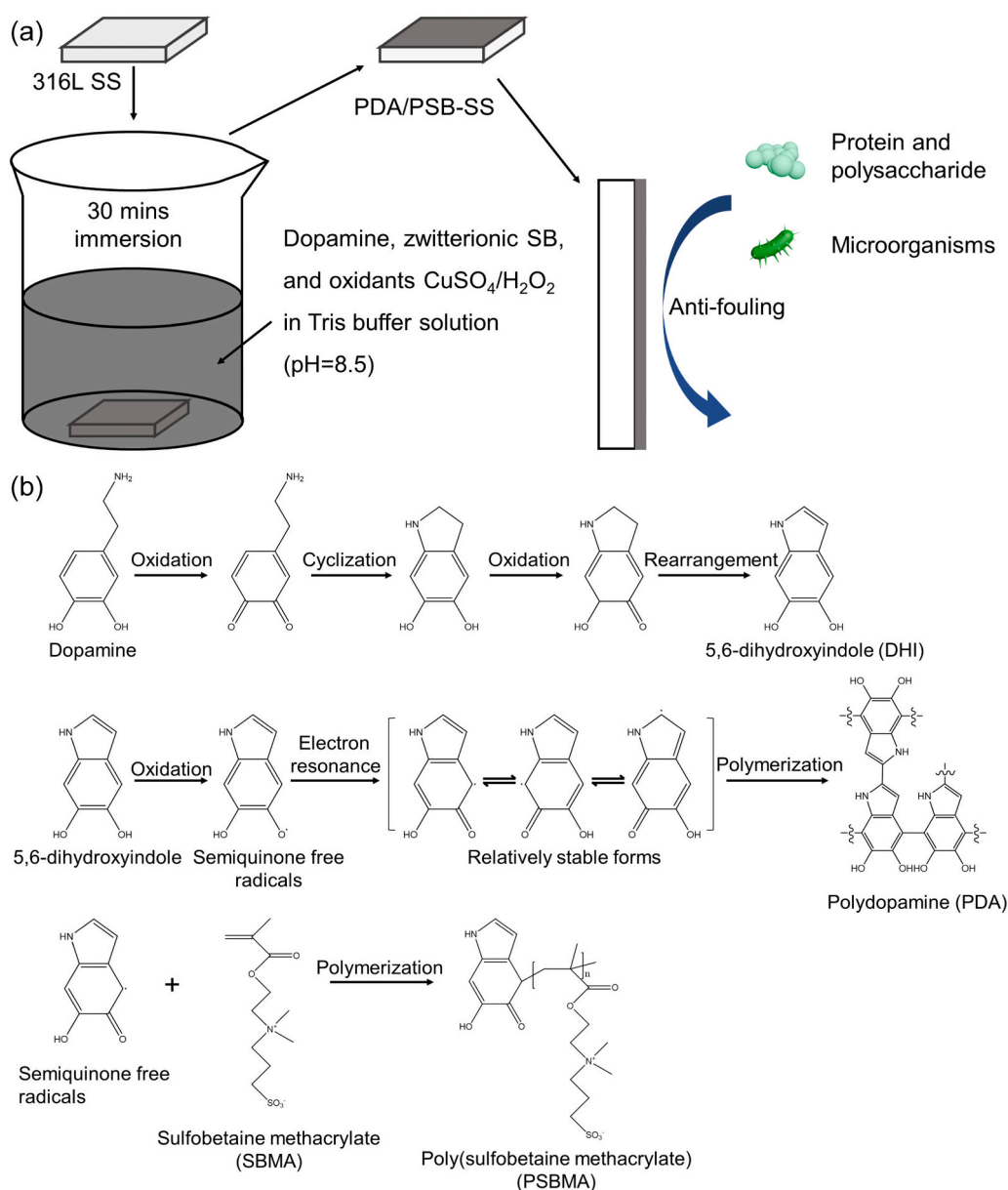


Fig. 1. (a) Schematic illustration for the fabrication of zwitterionic PDA/PSB coating on 316L SS surface; (b) possible mechanism for the dopamine-triggered simultaneous polymerization of dopamine and SB.

2. Experimental section

2.1. Materials and reagents

316 L SS was obtained from Biaozhilong Metal Products Co., Ltd. (Wuxi, China). Dopamine hydrochloride (98%) and [2-(methacryloyloxy)ethyl]-dimethyl (3-sulfopropyl) ammonium hydroxide (97%, SB) were purchased from Sigma-Aldrich. Tris(hydroxymethyl) aminomethane (Tris, 99%), hydrochloric acid (HCl, 35%), copper(II) sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 99%), hydrogen peroxide (H_2O_2 , 30%), bovine serum albumin (BSA), sodium alginate, and absolute ethanol were purchased from Sinopharm Chemical Reagent Co., Ltd. Fluorescent dye SYBR Green I and fluorescein isothiocyanate (FITC) were purchased from Solarbio Science & Technology Co., Ltd. (Beijing, China). Calcofluor stain was purchased from Maokang Biotechnology Co., Ltd. (Shanghai, China).

2.2. Fabrication of PDA/PSB coating on 316L SS surface

The fabrication process of PDA/PSB coating is illustrated in Fig. 1a. 316L SS samples were first ground using 240 #, 400 #, 800 #, 1200 #, 2000 # sandpapers in succession, and ultrasonically cleaned in absolute ethanol for 10 mins. The pretreated SS samples were placed into a mixed solution, i.e. a Tris buffer solution (50 mM, pH = 8.5) with dopamine (2 mg/mL) and SB (50 mg/mL). Then, CuSO_4 and H_2O_2 were added to the solution as oxidants, and final concentrations are 0.8 mg/mL and 0.67 mg/mL, respectively. After 30 mins of immersion, the samples were taken out, rinsed with Milli-Q water (18.2 M Ω), and dried with high-purity nitrogen gas (>99.99%).

2.3. Anti-fouling performance test

Organic adsorption assay was performed using BSA and sodium alginate, which are typical protein and polysaccharide, respectively. Each sample was pre-cleaned with absolute ethanol, and subsequently immersed in a BSA solution (5 mg/mL) or a sodium alginate solution (5 mg/mL) under static conditions for 24 h.

Microbial adhesion assay was performed using *Pseudomonas aeruginosa* (ATCC 9027) (Gram-negative bacterium), *Bacillus subtilis* (ATCC 6633) (Gram-positive bacterium) and *Navicula* sp. (diatom). Specifically, *P. aeruginosa* and *B. subtilis* were cultured in LB medium at 37 °C, and *Navicula* sp. was cultured in F/2 medium at 20 °C. When the microorganisms were cultured to their stationary phase, microbial cells were collected by centrifugation at 4000 rpm for 10 min at 4 °C. Then, the cell pellets were washed three times with sterile phosphate buffered saline (PBS, 10 mM, pH = 7.4) and resuspended in PBS. The final concentration of bacteria and diatom were 10^8 and 10^6 cells/mL, respectively. Each sample was pre-cleaned with absolute ethanol, and subsequently immersed in the cell suspension under static condition for 24 h.

2.4. Surface characterization

The chemical composition was confirmed by an X-ray photoelectron spectroscopy (XPS, Thermo Escalab 250Xi, USA) and a Fourier transform infrared spectrometer (FT-IR, Thermo Nico-let iN10, USA) with an attenuated total reflection (ATR) accessory. The hydrophilicity of different surfaces was assessed by measuring the water contact angles with an optical goniometer (Chenghui JGW 360A, China). After depositing a 5 μL water droplet on the sample surface, the shape of the droplet was recorded by a CCD camera, and the water contact angle was calculated by the accessorial software. The surface micromorphology was observed by a field emission scanning electron microscope (FE-SEM, FEI Quanta 250 FEG, USA) at 5.0 kV. The surface roughness was detected by an atomic force microscope (AFM, Bruker JPK NanoWizard 4, Germany) with a tapping probe (TESPA-V2, $k = 42 \text{ N/m}$).

A fluorescence microscope (Olympus DP74, Japan) was used to characterize the adsorption of BSA and sodium alginate, and adhesion of bacteria and diatoms. The sample surfaces adsorbed with BSA and sodium alginate were first stained with FITC for 20 mins and Calcofluor for 5 mins, respectively, in the dark, and then observed at an excitation wavelength of 495 nm and 355 nm, respectively. Before observation of bacteria and diatoms, the samples were gently rinsed with sterile PBS to remove unattached bacteria and diatoms. The adherent bacteria and diatoms were stained with SYBR Green I for 30 mins in dark, and then observed at an excitation wavelength of 497 nm. The density of adherent bacteria and diatoms was counted by ImageJ software [31]. The procedures are as follows: first import the image, then adjust the threshold to make the adherent cells (green fluorescent dots) to appear red, and finally apply the mathematical function “Analyze Particles” to the image. The results will be displayed in “Summary”, and the quantitative measurements were performed five times per group to obtain an average value. The data of adhesion cells were further analyzed by a professional statistical software, i.e. SPSS, and probability-value (p) is obtained by the significance test method, where, $p < 0.05$ and $p < 0.001$ are considered statistically significant and extremely significant, respectively.

3. Results and discussion

3.1. Fabrication mechanism of PDA/PSB coating

Currently, the polymerization mechanism of dopamine is not fully understood [32,33], while it has been confirmed that dopamine polymerization shares many characteristics with melanin biosynthesis, i.e. a free radical process [34]. The semiquinone free radicals generated during the dopamine oxidation can not only couple with each other to form PDA, but also initiate the polymerization of acrylate monomers [30,35]. Fig. 1b shows the possible mechanism for the dopamine-triggered simultaneous polymerization of dopamine and SB. Firstly, dopamine undergoes a series of oxidation, cyclization and molecular rearrangement to form 5,6-dihydroxyindole (DHI) under an alkaline condition [34]. Then, DHI is further oxidized to semiquinone free radicals, and subsequently transformed into relatively stable forms via electron resonance [36]. Finally, these free radicals are coupled to form PDA, or initiate the polymerization of zwitterionic SB monomers to form PSB.

Fig. 2a shows the color change of the mixed solution of dopamine and SB. The color of the solution quickly changes from colorless to pink exposed to air, indicating dopamine is easily to be oxidized under alkaline conditions. When CuSO_4 is added, the color immediately turns to green, and quickly changed from green to brown after adding H_2O_2 . This phenomenon suggests that the oxidative polymerization of dopamine is greatly accelerated by CuSO_4 and H_2O_2 . The color of the solution becomes darkness after 30 mins of reaction, i.e. a large amount of black PDA is generated in the mixed solution. Fig. 2b shows the digital photographs of different surfaces. Compared with SS surface, the surface of PDA-SS is the darkest, which is due to deposition of a certain amount of black PDA. The color of PDA/PSB-SS surface is lighter than PDA-SS surface, indicating that the amount of PDA on PDA/PSB-SS surface is less than PDA-SS surface. This is attributed to that the presence of SB consumes part of semiquinone free radicals during the process of polymerization. Therefore, this result further confirms that the polymerization of SB is initiated by the semiquinone free radicals produced by the oxidation of dopamine in alkaline solution.

3.2. Chemical composition of PDA/PSB coating

Figs. 3a and 3b show the survey XPS and S 2p spectra of the PDA-SS and PDA/PSB-SS surfaces, respectively. There are three obvious peaks in the XPS spectra of PDA-SS and PDA/PSB-SS, i.e. the photoelectron peaks at 284.5 eV, 532.5 eV and 400.0 eV correspond to the binding energy of

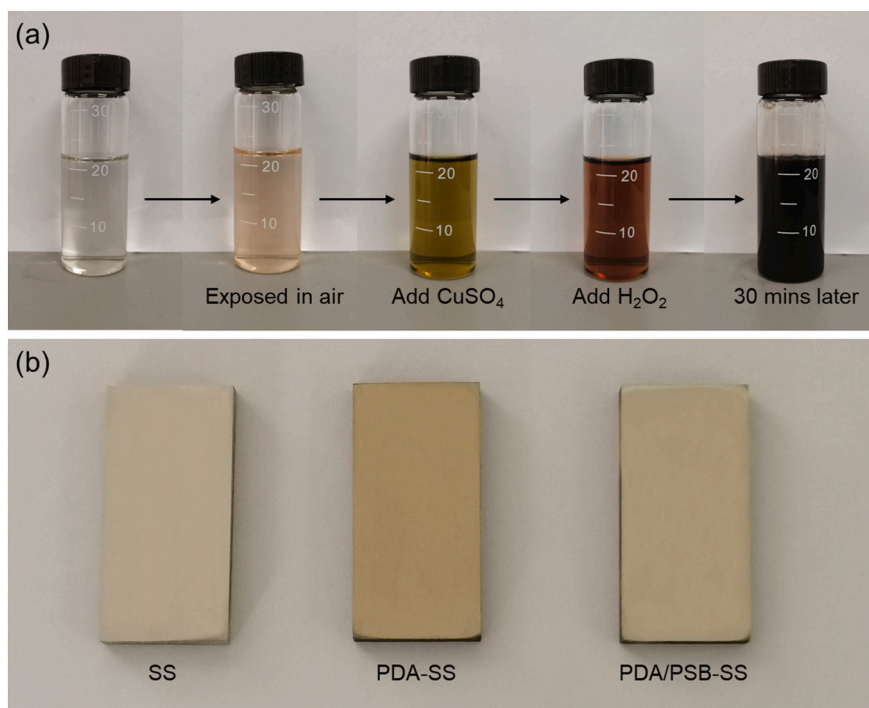


Fig. 2. Digital photographs of (a) the color change of the mixed solution of dopamine and SB, and (b) surface color of different samples.

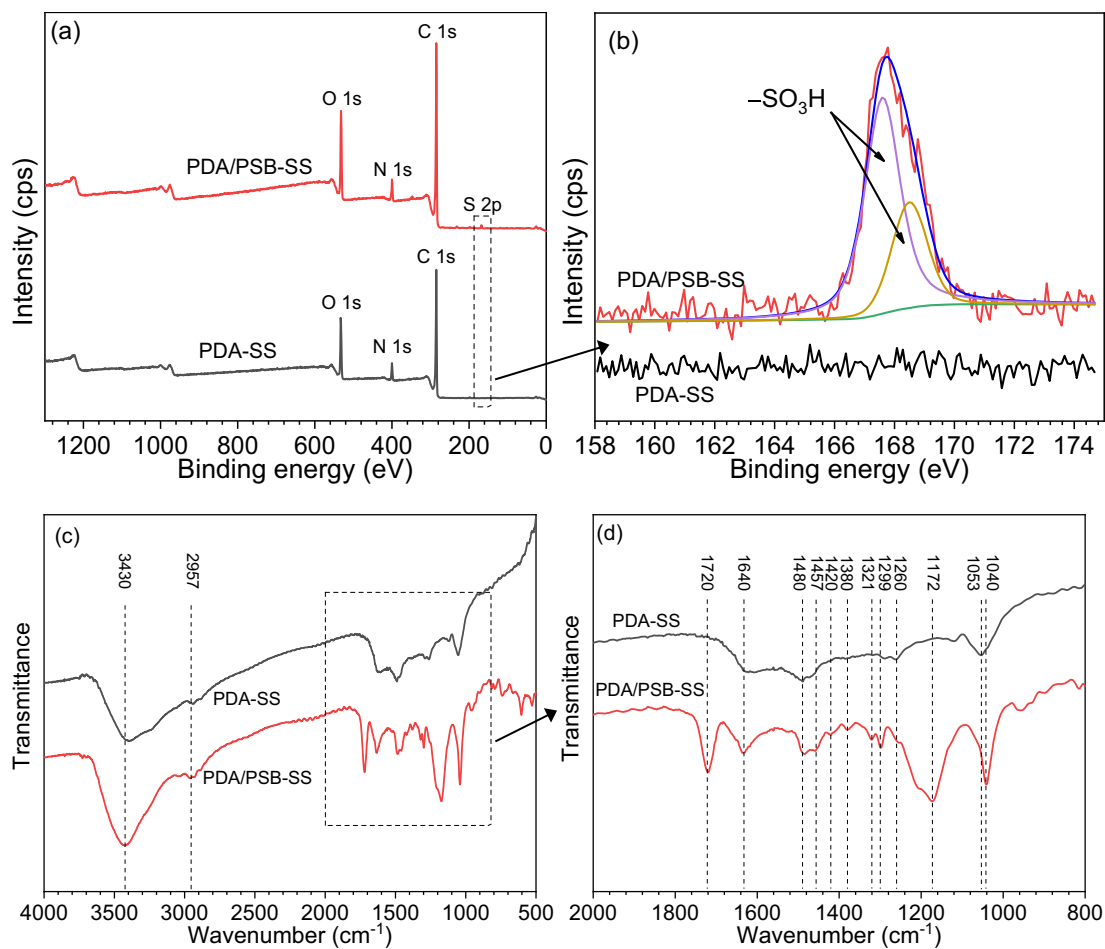


Fig. 3. (a) Survey XPS, (b) S 2p and (c, d) FTIR spectra of PDA/SS and PDA/PSB-SS surfaces.

C 1s, O 1s and N 1s, respectively (Fig. 3a). However, there is a new peak located at 167.78 eV in spectrum of PDA/PSB-SS (Fig. 3b), which is ascribed to the binding energy of S 2p. Normally, S 2p peak is fitted with a doublet representing spin orbit splitting of S 2p_{3/2} (167.58 eV) and S 2p_{1/2} (168.48 eV) peaks, which are ascribed to sulfonic acid group ($-\text{SO}_3\text{H}$) [37]. According to the molecular formulas of PDA and PSB, this result indicates the presence of PDA on both of the surfaces and zwitterionic polymer PSB on the PDA/PSB-SS surface.

Figs. 3c and 3d show the FTIR spectra of the PDA-SS and PDA/PSB-SS surfaces. The characteristic peaks of PDA are observed in FTIR spectrum of PDA-SS. The wide peaks round 3430 cm^{-1} and 2957 cm^{-1} are caused by stretching vibrations of X—H (X: O and N) and C—H, respectively [38]. The peaks at 1640 cm^{-1} and 1053 cm^{-1} correspond to the deformation vibration of N—H and stretching vibration of C—O, respectively. The peaks at 1480 cm^{-1} and 1457 cm^{-1} are attributed to stretching vibrations of C=C and the asymmetric stretching of C—H, respectively [39]. A double peak at 1286 cm^{-1} and 1260 cm^{-1} is associated with stretching vibrations of C—N [38]. Besides above peaks there are some new peaks in FTIR spectrum of PDA/PSB-SS surface. The peaks at 1720 cm^{-1} correspond to the stretching vibration of C=O. The peaks at 1420 cm^{-1} and 1380 cm^{-1} is associated with deformation vibration of C—H. A double peak at 1321 cm^{-1} and 1299 cm^{-1} is associated with the deformation vibration of N—H [40]. The 1040 cm^{-1} and 1172 cm^{-1} bands can be assigned to sulfonic acid groups [41]. These peaks indicate the existence of PSB on the PDA/PSB-SS surface, which is consistent with the results of XPS.

3.3. Surface hydrophilicity and micromorphology

Fig. 4 shows the static water contact angles of different surfaces and PDA/PSB-SS surfaces at different time. Compared with the water contact angle on unmodified SS surface, i.e. $49.7 \pm 7.2^\circ$, the surface hydrophilicity of PDA-SS is improved, i.e. the water contact angle is decreased to $21.5 \pm 2.8^\circ$ (Fig. 4a). This is attributed to that the large number of hydroxyl groups in PDA can bind water molecule via hydrogen bonding. However, the PDA/PSB SS surface becomes super-hydrophilic, i.e. the water contact angle further decreases to $7.3 \pm 1.8^\circ$ (Fig. 4a). This is due to that the presence of zwitterionic PSB binds water molecule more strongly via electrostatic hydration. Additionally, it can be seen that the hydrophilicity of the PDA/PSB coating does not significantly change with the time, i.e. the water contact angle is around 7° in 1 week (Fig. 4b). This behavior indicates the PDA/PSB coating has a good stability.

Figs. 5 and 6 present the micromorphology of PDA-SS and PDA/PSB-SS surfaces. As shown in Fig. 5a, many particulate aggregates with a diameter of about 130 nm are observed on the PDA-SS surface, which agrees with the result of previous study [42], i.e. the polymerization of

dopamine generates many melanin-like nanoparticles. However, the nanoparticles on PDA/PSB-SS surface are more and smaller (with a diameter of about 40 nm) (Fig. 5b). This may be due to that the presence of SB monomers competes with dopamine for free radicals in the polymerization system, which inhibits the formation of large particulate aggregates of PDA. The results of AFM show that the PDA/PSB-SS surface is smoother than PDA-SS surface, i.e. the root-mean-square roughness (R_q) of PDA-SS and PDA/PSB-SS surfaces is 6.644 nm and 3.860 nm, respectively, and the maximum vertical height is 29.2 nm and 17.3 nm, respectively (Fig. 6). This attributes to that the nanoparticles in PDA/PSB coating are smaller and distribute more evenly than those in PDA coating, which is in good agreement with the observation of SEM.

3.4. Anti-fouling performance of PDA/PSB coating

The conditioning film formed by accumulation of organic molecules provides nutritional conditions for subsequent microbial attachment and growth, which is the first step of marine biofouling [43,44]. Normally, proteins and polysaccharides are the main organic molecules in conditioning film. In this work, BSA and sodium alginate were selected on behalf of proteins and polysaccharides, respectively, to evaluate the adsorption resistance of PDA/PSB coating. Fig. 7 shows the fluorescent images of BSA and sodium alginate adsorbed on different surfaces. The green and blue signals represent FITC-labeled BSA and Calcofluor-labeled sodium alginate, respectively. Plenty of BSA and sodium alginate adsorb on the unmodified SS surface (Figs. 7a and 7d), indicating SS has a strong chelating effect with protein and polysaccharides. Compared with the SS surface, the amounts of both BSA and sodium alginate decrease on the PDA-SS surface (Figs. 7b and 7e), which means that both of protein and polysaccharide adsorb onto SS surface more easily than PDA-SS surface. However, there is almost no BSA and sodium alginate on the PDA/PSB-SS surface (Figs. 7c and 7f), which suggests that the PDA/PSB coating can effectively inhibit the nonspecific organic molecules adsorption owing to its super-hydrophilicity.

Microbial attachment has been considered as one of the critical steps of marine biofouling [44]. Herein, three typical marine microorganisms, i.e. Gram-negative bacterium (*P. aeruginosa*), Gram-positive bacterium (*B. subtilis*) and diatom (*Navicula* sp.), are chosen to evaluate the microbial resistance of PDA/PSB coating. Figs. 8 and 9 show the fluorescent images of microbial adhesion and statistical results of adherent cell density, respectively. Similar to the results in Fig. 7, numerous cells of *P. aeruginosa*, *B. subtilis* and *Navicula* sp. adhere on the unmodified SS surface, i.e. the numbers of adherent *P. aeruginosa*, *B. subtilis* and *Navicula* sp. are about 949, 1262 and 848 cells/ mm^2 , respectively (Figs. 8a, 8d and 8g). This result indicates that the unmodified SS surface is susceptible to microbial adhesion and developing into biofouling. The adhesion density of *B. subtilis* is significantly higher than that of

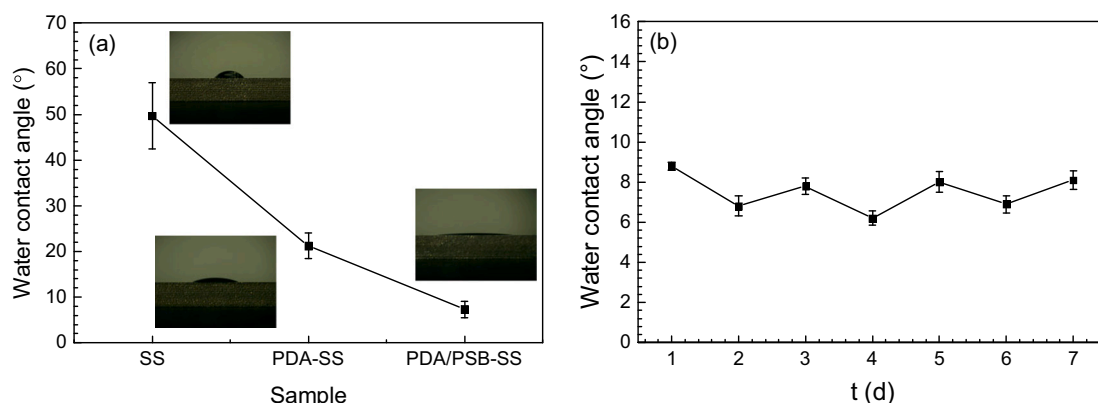


Fig. 4. Water contact angles of (a) different surfaces and (b) PDA/PSB-SS surfaces at different time. Each point is the average of three measurements on three replicates, and error bars show the standard deviation.

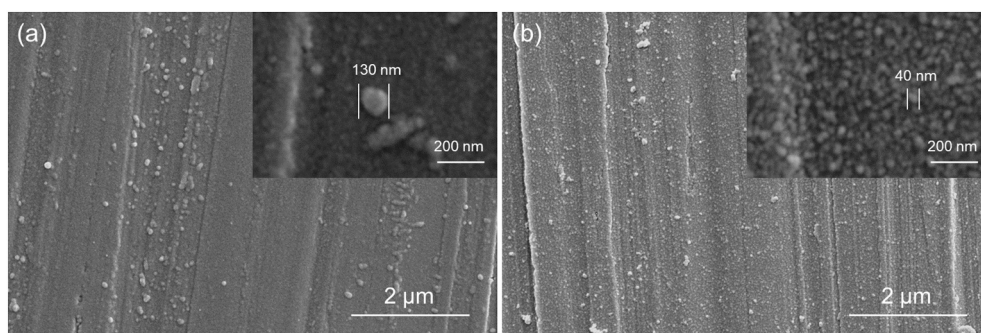


Fig. 5. SEM images of (a) PDA-SS and (b) PDA/PSB-SS surfaces.

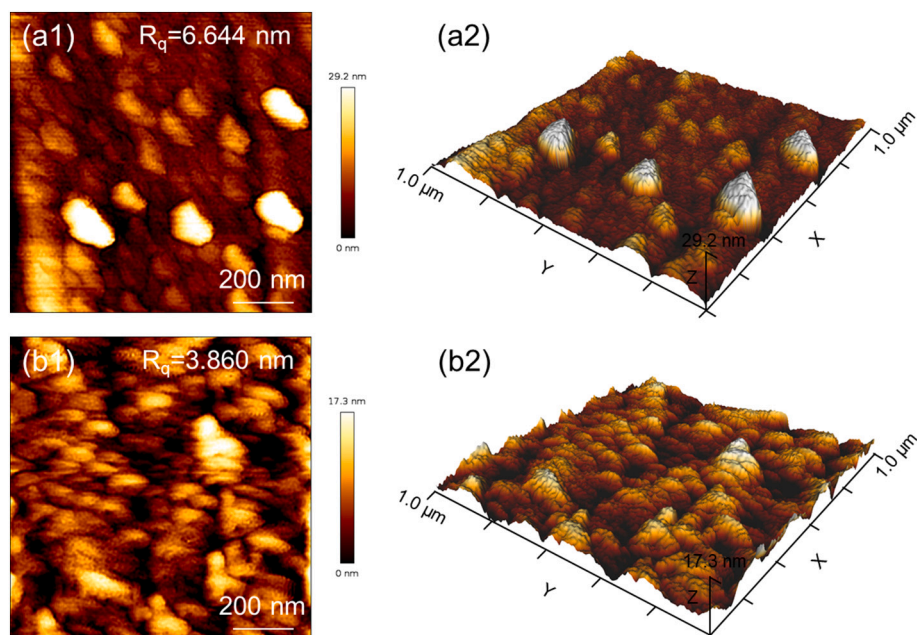


Fig. 6. AFM images of (a) PDA-SS and (b) PDA/PSB-SS surfaces.

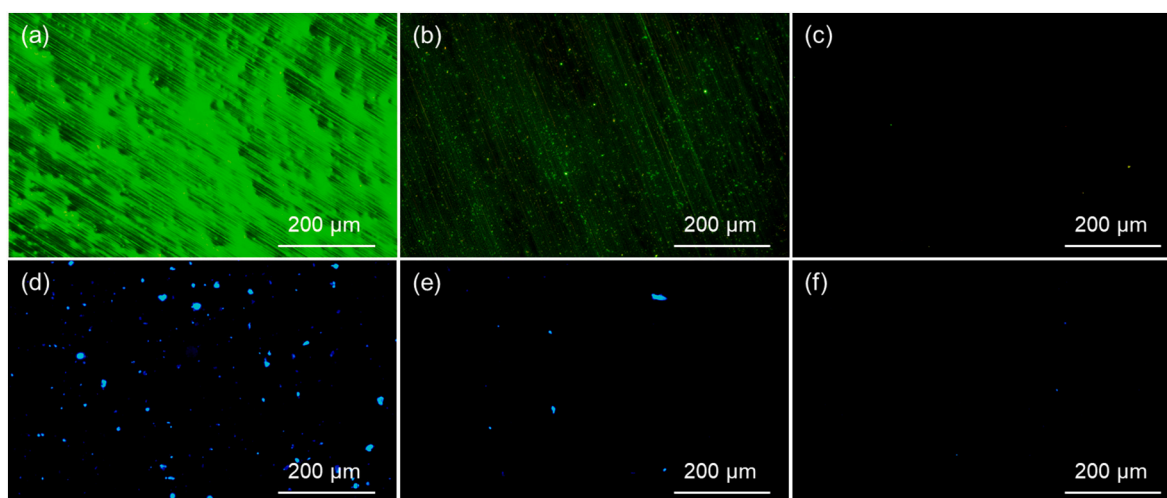


Fig. 7. Fluorescent images of (a–c) BSA and (d–f) sodium alginate adsorbed on the surfaces of (a, d) SS, (b, e) PDA-SS and (c, f) PDA/PSB-SS after 24 h immersion. The green signals are FITC-labeled BSA, and the blue signals are Calcofluor-labeled sodium alginate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

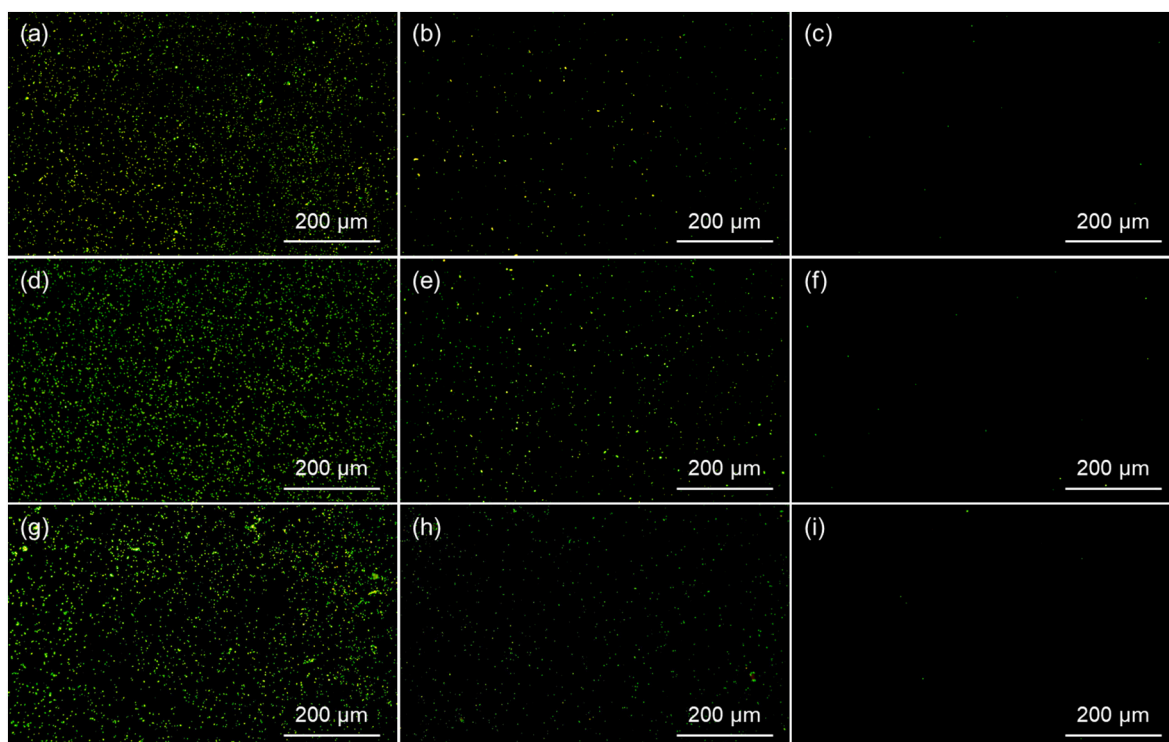


Fig. 8. Fluorescent images of (a–c) *P. aeruginosa*, (d–f) *B. subtilis* and (g–i) *Navicula* sp. adhered on the surfaces of (a, d, g) SS, (b, e, h) PDA-SS and (c, f, i) PDA/PSB-SS after 24 h immersion. The green signals are microbial cells stained by SYBR Green I. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

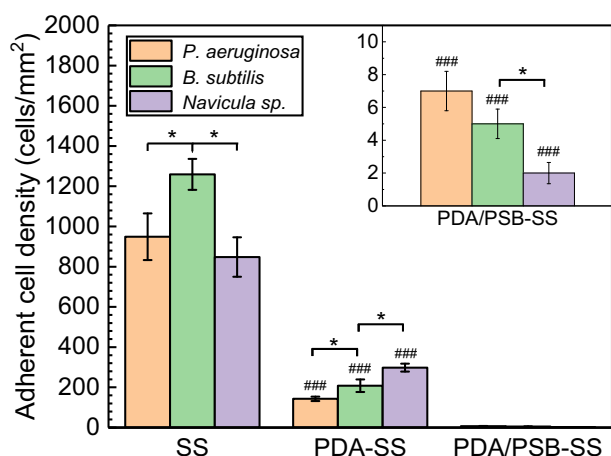


Fig. 9. Adherent cell density of *P. aeruginosa*, *B. subtilis* and *Navicula* sp. on different surfaces. Each datum is the average of five measurements ($n = 4$), and error bars show the standard deviation. * represents p -value < 0.05 , ### represents p -value < 0.001 as compared with SS surface as the control.

P. aeruginosa and *Navicula* sp. ($p < 0.05$) on the unmodified SS surface (Fig. 9), which means that *B. subtilis* adheres more easily than *P. aeruginosa* and *Navicula* sp. onto SS surface. The numbers of adherent *P. aeruginosa*, *B. subtilis* and *Navicula* sp. on the PDA-SS surface decrease obviously (Figs. 8b, 8e and 8h), i.e. they are about 143, 209 and 298 cells/mm², respectively (Fig. 9). The adhesion densities of *P. aeruginosa*, *B. subtilis* and *Navicula* sp. ($p < 0.001$) are reduced by about 84.9%, 83.4% and 64.9%, respectively. This result indicates that the PDA coating has a certain degree of performance in inhibiting microbial adhesion. By contrast, after modifying a layer of PDA/PSB coating on the SS surface, the adhesion densities of the three microbial species are

decreased even more ($p < 0.001$), and only very few microbial cells can be observed (Figs. 8c, 8f and 8i). The numbers of adherent *P. aeruginosa*, *B. subtilis* and *Navicula* sp. are about 7, 5 and 2 cells/mm², respectively, on the PDA/PSB-SS surface (Fig. 9), i.e., the adhesion densities of the three microbial species are reduced by more than 99%. This result indicates that the PDA/PSB coating can effectively inhibit microbial adhesion. Due to the presence of zwitterionic polymer PSB, the PDA/PSB coating is super-hydrophilic, and can form a hydration layer to prevent the direct contact between microorganisms and surface [12]. In view of the key role of organic molecules adsorption and microbial adhesion in the formation of marine biofouling, it can be inferred that the PDA/PSB coating has the function of slowing down the development of marine biofouling.

4. Conclusion

In summary, zwitterionic PDA/PSB coating is fabricated on 316L SS surface via one-step simultaneous polymerization and codeposition of dopamine and SB within 30 min. The polymerization of SB is initiated by the semiquinone free radicals produced by the oxidation of dopamine in alkaline solution. The presence of PSB results in super-hydrophilicity on surface of PDA/PSB-SS. The PDA/PSB-SS can inhibit the adsorption of organic molecules, i.e. BSA and sodium alginate. The adhesions of *P. aeruginosa*, *B. subtilis* and *Navicula* sp. are reduced by more than 99% after 24 h of exposure. The first step of marine biofouling is controlled by PDA/PSB coating on 316L SS surface.

CRediT authorship contribution statement

Jiashun Shi: Conceptualization, Methodology, Data curation, Writing – original draft. **Yudi Zhao:** Investigation, Data curation. **Weigang Wang:** Investigation, Data curation. **Huimin Liao:** Formal analysis. **Guangzhou Liu:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] J.A. Callow, M.E. Callow, Trends in the development of environmentally friendly fouling-resistant marine coatings, *Nat. Commun.* 2 (2011) 1–10.
- [2] M.P. Schultz, J.A. Bendick, E.R. Holm, W.M. Hertel, Economic impact of biofouling on a naval surface ship, *Biofouling* 27 (2011) 87–98.
- [3] M. Lejars, A. Margailan, C. Bressy, Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings, *Chem. Rev.* 112 (2012) 4347–4390.
- [4] D.M. Yebra, S. Kiil, K. Dam-Johansen, Antifouling technology - past, present and future steps towards efficient and environmentally friendly antifouling coatings, *Prog. Org. Coat.* 50 (2004) 75–104.
- [5] C.M. Kirschner, A.B. Brennan, Bio-inspired antifouling strategies, *Annu. Rev. Mater. Res.* 42 (2012) 211–229.
- [6] G. Cheng, Z. Zhang, S. Chen, J.D. Bryers, S. Jiang, Inhibition of bacterial adhesion and biofilm formation on zwitterionic surfaces, *Biomaterials* 28 (2007) 4192–4199.
- [7] F. Saffarimandoab, B.Y. Gul, S. Erkoc-Ilter, S. Guclu, S. Unal, B. Tunaboylu, Y. Z. Menciloglu, I. Koyuncu, Evaluation of biofouling behavior of zwitterionic silane coated reverse osmosis membranes fouled by marine bacteria, *Prog. Org. Coat.* 134 (2019) 303–311.
- [8] Y. Zhu, X. Yu, T. Zhang, X. Wang, Constructing zwitterionic coatings on thin-film nanofibrous composite membrane substrate for multifunctionality, *Appl. Surf. Sci.* 483 (2019) 979–990.
- [9] Q. Ye, B. He, Y. Zhang, J. Zhang, S. Liu, F. Zhou, Grafting robust thick zwitterionic polymer brushes via subsurface-initiated ring-opening metathesis polymerization for antimicrobial and anti-biofouling, *ACS Appl. Mater. Interfaces* 11 (2019) 39171–39178.
- [10] G. Cheng, G. Li, H. Xue, S. Chen, J.D. Bryers, S. Jiang, Zwitterionic carboxybetaine polymer surfaces and their resistance to long-term biofilm formation, *Biomaterials* 30 (2009) 5234–5240.
- [11] L. Zhang, Z. Cao, T. Bai, L. Carr, J.R. Ella-Menye, C. Irvin, B.D. Ratner, S. Jiang, Zwitterionic hydrogels implanted in mice resist the foreign-body reaction, *Nat. Biotechnol.* 31 (2013) 553–556.
- [12] D. Campoccia, L. Montanaro, C.R. Arciola, A review of the biomaterials technologies for infection-resistant surfaces, *Biomaterials* 34 (2013) 8533–8554.
- [13] S. Chen, L. Li, C. Zhao, J. Zheng, Surface hydration: principles and applications toward low-fouling/nonfouling biomaterials, *Polymer* 51 (2010) 5283–5293.
- [14] M. Krishnamoorthy, S. Hakobyan, M. Ramstedt, J.E. Gautrot, Surface-initiated polymer brushes in the biomedical field: applications in membrane science, biosensing, cell culture, regenerative medicine and antibacterial coatings, *Chem. Rev.* 114 (2014) 10976–11026.
- [15] S. Jiang, Z. Cao, Ultralow-fouling, functionalizable, and hydrolyzable zwitterionic materials and their derivatives for biological applications, *Adv. Mater.* 22 (2010) 920–932.
- [16] Z. Zhang, S. Chen, Y. Chang, S. Jiang, Surface grafted sulfobetaine polymers via atom transfer radical polymerization as superlow fouling coatings, *J. Phys. Chem. B* 110 (2006) 10799–10804.
- [17] H.S. Sundaram, X. Han, A.K. Nowinski, J.R. Ella-Menye, C. Wimbish, P. Marek, K. Senecal, S. Jiang, One-step dip coating of zwitterionic sulfobetaine polymers on hydrophobic and hydrophilic surfaces, *ACS Appl. Mater. Interfaces* 6 (2014) 6664–6671.
- [18] Z. Zhang, J.A. Finlay, L. Wang, Y. Gao, J.A. Callow, M.E. Callow, S. Jiang, Polysulfobetaine-grafted surfaces as environmentally benign ultralow fouling marine coatings, *Langmuir* 25 (2009) 13516–13521.
- [19] Z. Zhang, T. Chao, S. Chen, S. Jiang, Superlow fouling sulfobetaine and carboxybetaine polymers on glass slides, *Langmuir* 22 (2006) 10072–10077.
- [20] H. Lee, S.M. Dellatore, W.M. Miller, P.B. Messersmith, Mussel-inspired surface chemistry for multifunctional coatings, *Science* 318 (2007) 426–430.
- [21] Q. Wei, F. Zhang, J. Li, B. Li, C. Zhao, Oxidant-induced dopamine polymerization for multifunctional coatings, *Polym. Chem.* 1 (2010) 1430–1433.
- [22] Y. Liu, K. Ai, L. Lu, Polydopamine and its derivative materials: synthesis and promising applications in energy, environmental, and biomedical fields, *Chem. Rev.* 114 (2014) 5057–5115.
- [23] C. Zhang, Y. Ou, W.X. Lei, L.S. Wan, J. Ji, Z.K. Xu, CuSO₄/H₂O₂-induced rapid deposition of polydopamine coatings with high uniformity and enhanced stability, *Angew. Chem. Int. Edit.* 55 (2016) 3054–3057.
- [24] R. Zhou, P. Ren, H. Yang, Z. Xu, Fabrication of antifouling membrane surface by poly(sulfobetaine methacrylate)/polydopamine co-deposition, *J. Membr. Sci.* 466 (2014) 18–25.
- [25] B. Wang, Y. Quan, Z. Xu, Q. Zhao, Preparation of highly effective antibacterial coating with polydopamine/chitosan/silver nanoparticles via simple immersion, *Prog. Org. Coat.* 149 (2020), 105967.
- [26] J. Xu, N. Xu, T. Zhou, X. Xiao, B. Gao, J. Fu, T. Zhang, Polydopamine coatings embedded with silver nanoparticles on nanostructured titania for long-lasting antibacterial effect, *Surf. Coat. Technol.* 320 (2017) 608–613.
- [27] P. Cao, C. Du, X. He, C. Zhang, C. Yuan, Modification of a derived antimicrobial peptide on steel surface for marine bacterial resistance, *Appl. Surf. Sci.* 510 (2020), 145512.
- [28] P. Cao, K. Liu, X. Liu, W. Sun, D. Wu, C. Yuan, X. Bai, C. Zhang, Antibacterial properties of Magainin II peptide onto 304 stainless steel surfaces: a comparison study of two dopamine modification methods, *Colloid. Surface B* 194 (2020), 111198.
- [29] Y. Qiao, Y. Li, Q. Zhang, Q. Wang, J. Gao, L. Wang, Dopamine-mediated zwitterionic polyelectrolyte-coated polypropylene hernia mesh with synergistic anti-inflammation effects, *Langmuir* 36 (2020) 5251–5261.
- [30] C. Zhang, M. Ma, T. Chen, H. Zhang, D. Hu, B. Wu, J. Ji, Z. Xu, Dopamine-triggered one-step polymerization and codeposition of acrylate monomers for functional coatings, *ACS Appl. Mater. Interfaces* 9 (2017) 34356–34366.
- [31] C.A. Schneider, W.S. Rasband, K.W. Eliceiri, NIH image to ImageJ: 25 years of image analysis, *Nat. Methods* 9 (2012) 671–675.
- [32] J. Liebscher, R. Mrowczynski, H.A. Scheidt, C. Filip, N.D. Hadade, R. Turcu, A. Bende, S. Beck, Structure of polydopamine: a never-ending story? *Langmuir* 29 (2013) 10539–10548.
- [33] D.R. Dreyer, D.J. Miller, B.D. Freeman, D.R. Paul, C.W. Bielawski, Elucidating the structure of poly(dopamine), *Langmuir* 28 (2012) 6428–6435.
- [34] S. Hong, Y.S. Na, S. Choi, I.T. Song, W.Y. Kim, H. Lee, Non-covalent self-assembly and covalent polymerization co-contribute to polydopamine formation, *Adv. Funct. Mater.* 22 (2012) 4711–4717.
- [35] M. Ma, C. Zhang, T. Chen, J. Yang, J. Wang, J. Ji, Z. Xu, Bioinspired polydopamine/polyzwitterion coatings for underwater anti-oil and -freezing surfaces, *Langmuir* 35 (2019) 1895–1901.
- [36] C.T. Chen, F.J. Martin-Martinez, G.S. Jung, M.J. Buehler, Polydopamine and eumelanin molecular structures investigated with *ab initio* calculations, *Chem. Sci.* 8 (2017) 1631–1641.
- [37] N. Noor, J. Koll, N. Scharnagl, C. Abetz, V. Abetz, Hollow fiber membranes of blends of polyethersulfone and sulfonated polymers, *Membranes* 8 (2018) 54.
- [38] L. Wang, L. Yan, X. Gao, Effect of electron beam irradiation on polydopamine and its application in polymer solar cells, *Int. J. Energy Res.* 42 (2018) 3496–3505.
- [39] T. Fatima, R.A. Haque, M.R. Razali, A. Ahmad, M.A. Iqbal, M. Asif, M.B.K. Ahamed, A.M.S. Abdul Majid, Synthesis, crystal structure, in vitro anticancer and in vivo acute oral toxicity studies of tetramethylene linked bis-benzimidazolium salts and their respective dinuclear Ag (I)-NHC complexes, *J. Coord. Chem.* 69 (2016) 3367–3383.
- [40] H. Wang, Z.W. Zhai, Y.X. Shi, C.X. Tan, J.Q. Weng, L. Han, B.J. Li, X.H. Liu, Novel trifluoromethylpyrazole acyl thiourea derivatives: synthesis, antifungal activity and docking study, *Lett. Drug Des. Discov.* 16 (2019) 785–791.
- [41] A. Shavandi, A. Carne, A.A. Bekhit, A.E.D.A. Bekhit, An improved method for solubilisation of wool keratin using peracetic acid, *J. Environ. Chem. Eng.* 5 (2017) 1977–1984.
- [42] K.Y. Ju, Y. Lee, S. Lee, S.B. Park, J.K. Lee, Bioinspired polymerization of dopamine to generate melanin-like nanoparticles having an excellent free-radical-scavenging property, *Biomacromolecules* 12 (2011) 625–632.
- [43] A. Jain, N.B. Bhosle, Biochemical composition of the marine conditioning film: implications for bacterial adhesion, *Biofouling* 25 (2009) 13–19.
- [44] A.J. de Kerchove, M. Elimelech, Impact of alginate conditioning film on deposition kinetics of motile and nonmotile *Pseudomonas aeruginosa* strains, *Appl. Environ. Microbiol.* 73 (2007) 5227–5234.